Question #1 of 111

Question ID: 1473752

Which of the following statements regarding an option's price is CORRECT? An option's price is:

A) an increasing function of the underlying asset's volatility.

B) a decreasing function of the underlying asset's volatility.

C) a decreasing function of the underlying asset's volatility when it has a long time remaining until expiration and an increasing function of its volatility if the option is close to expiration.

Question #2 of 111

Question ID: 1473755

Two call options have the same delta but option A has a higher gamma than option B. When the price of the underlying asset increases, the number of option A calls necessary to hedge the price risk in 100 shares of stock, compared to the number of option B calls, is a:

A) larger positive number.

B) larger (negative) number.

C) smaller (negative) number.

Question #3 of 111

Question ID: 1473748

How is the gamma of an option defined? Gamma is the change in the:

A) delta as the price of the underlying security changes.

B) vega as the option price changes.

C) option price as the underlying security changes.

Question #4 of 111

Question ID: 1473685

Referring to put-call parity, which one of the following alternatives would allow you to create a synthetic stock position?

A) Buy a European call option; buy a European put option; invest the present value of the exercise price in a riskless pure-discount bond.

B) Buy a European call option; short a European put option; invest the present value of the exercise price in a riskless pure-discount bond.

C) Sell a European call option; buy a European put option; short the present value of the exercise price worth of a riskless pure-discount bond.

Question #5 of 111

Question ID: 1473743

The value of a put option is positively related to all of the following EXCEPT:

A) risk-free rate.

B) exercise price.

C) time to maturity.

Max Perrot, CFA, works for WWF, a mortgage banking company which originates residential mortgage loans. On a monthly basis, WWF issues agency mortgage-backed securities (MBS) backed by their loans. WWF sells the MBS in the open market soon after securitization, but retains the servicing rights to the loans. WWF currently owns the third largest mortgage servicing portfolio in the U.S. Perrot has recently been promoted to Senior Vice President of Asset and Liability Management for WWF. Perrot's new responsibilities encompass hedging WWF's newly created MBS prior to their sale, as well as managing the interest rate exposure on the servicing portfolio. Both types of assets are extremely sensitive to changes in interest rates, though not necessarily in the same manner.

Although WWF has retained all of the servicing rights of its loans in the past, they are not opposed to the selling of portions of the portfolio if market conditions are right. WWF's management wants Perrot in his new position to focus primarily on preserving the value of the servicing portfolio through hedging strategies that are cost effective to execute. Also, any hedge strategy used by Perrot must be extremely liquid in the event that a portion of the servicing portfolio is sold and the hedge needs to be unwound. The upper management of WWF anticipates a period of volatility in interest rates, and they have asked Perrot to project expected returns of a hedged position under a variety of interest rates scenarios.

Perrot's predecessor lacked experience in hedging with swaps and futures contracts, but he had used them periodically with lackluster results. Through his inaction, he had exposed the firm to significant asset and liability mismatch, which had increased dramatically over the past two years as both production and the servicing portfolio had grown. Perrot, on the other hand, had extensive experience with hedging with derivatives in his prior job. He is familiar with executing hedging strategies utilizing not only swap and futures, but also with options such as caps and floors. He decides that before he presents any potential hedging strategy to WWF's management, he would first like to bring them up to speed on the basic hedging concepts. He prepares a brief presentation on the relationships between interest rates and options, and outlines some basic hedging strategies. He anticipates many questions that may arise from his presentation, and prepares a handout in a question and answer format.

Question #6 - 9 of 111

Question ID: 1586311

Assume that a three-year semi-annually settled floor with a strike rate of 8% and a notional amount of $100 million is being analyzed. The reference rate is six-month London Interbank Offered Rate (LIBOR). Suppose that LIBOR for the next four semi-annual periods is as follows:

Period	LIBOR
1	7.5%
2	8.2%
3	8.1%
4	8.7%

What is the payoff for the floor for period 1?

A) $0.

B) $250,000.

C) $500,000.

Question #7 - 9 of 111

Question ID: 1586312

Which of the following *best* explains the difference between an interest rate put option and a put option on a fixed income security? The interest rate put option value:

A) decreases if interest rates increase just as the value of a put option on a fixed income security decreases.

B) decreases if interest rates increase while the value of a put option on a fixed income security increases if interest rates increase.

C) increases if interest rates increase just as the value of a put option on a fixed income security increases.

Question #8 - 9 of 111

Question ID: 1586313

A LIBOR based floating rate bond combined with a LIBOR based collar (a short position in an interest rate cap and a long position in an interest rate floor both at the same strike rate) is equivalent to a:

A) pay-fixed swap position.

B) fixed-rate bond.

C) call option on a bond.

Question #9 - 9 of 111

Question ID: 1586314

Which of the following is *most likely* a reason why dynamic riskless arbitrage is difficult in real markets?

A) Securities are subject to insider trading.

B) Continuous rebalancing.

C) Short sale constraints exist.

You are interested in derivative products, particularly with a view to identifying arbitrage opportunities. You start with bond futures:

- The cheapest to deliver (CTD) bond underlying the T-bond futures contract maturing in five months is a 4.6% T-Bond currently priced at $1,002.33 (full price) per $1,000 par. The CTD paid its last coupon four months ago, and its conversion factor is 1.13. The risk free rate is 2.99%.

Peter Wang, one of your colleague, knew of your interest in derivative products advises you to consider interest rate options and swaptions. Wang makes the following comments:

Comment 1:	An investor having a long position in a call option on a bond has the same position as if he is long an interest rate floor.
Comment 2:	A borrower of a floating rate loan can create an interest rate collar by buying an interest rate cap and selling an interest rate floor and the cap sets the maximum interest rate payable by the borrower.
Comment 3:	A payer swaption is the right to enter into a specific swap at some date in the future as the fixed-rate payer. A payer swaption becomes more valuable if an equivalent swap at the market rate is higher than the strike rate.

Question #10 - 13 of 111

Question ID: 1473723

Which of the following is *closest* to the no-arbitrage price of the 5-month T-Bond futures contract?

A) $867.20.

B) $877.47.

C) $976.02.

Question #11 - 13 of 111

Question ID: 1473724

Comment 1 is *best* described as:

A) correct.

B) incorrect as long an interest rate floor should be short an interest rate floor.

C) incorrect as long an interest rate floor should be long an interest rate cap.

Question #12 - 13 of 111

Question ID: 1473725

Comment 2 is *best* described as:

A) correct.

B) incorrect as it should be buying an interest rate floor and selling an interest rate cap.

C) incorrect as it should be the floor that sets the maximum interest rate payable by the borrower.

Question #13 - 13 of 111

Question ID: 1473726

Comment 3 can be *best* described as:

A) correct.

B) incorrect as it is describing a receiver swaption, not a payer swaption.

C) incorrect as a payer swaption is more valuable if an equivalent swap at the market rate is lower than the strike rate.

Question #14 of 111

Question ID: 1473753

Compared to the delta of a long position in a stock, the delta of an at-the-money call option on the stock is *most likely* to be:

A) the same.

B) less.

C) greater.

Question #15 of 111

Question ID: 1473705

Which of the following is NOT one of the assumptions of the Black-Scholes-Merton option-pricing model?

A) The yield curve for risk-free assets is fixed over the term of the option.

B) There are no taxes and transactions costs are zero for options and arbitrage portfolios.

C) Early exercise is not allowed.

Frank Potter, CFA, a financial adviser for Star Financial, LLC has been hired by John Williamson, a recently retired executive from Reston Industries. Over the years Williamson has accumulated $10 million worth of Reston stock and another $2 million in a cash savings account. Potter has a number of unconventional investment strategies for Williamson's portfolio; many of the strategies include the use of various equity derivatives.

Potter's first recommendation involves the use of a total return equity swap. Potter outlines the characteristics of the swap in Table 1. In addition to the equity swap, Potter explains to Williamson that there are numerous options available for him to obtain almost any risk return profile he might need. Potter suggest that Williamson consider options on both Reston stock and the S&P 500. Potter collects the information needed to evaluate options for each security. These results are presented in Table 2.

Table 1: Specification of Equity Swap

Term	3 years
Notional principal	$10 million
Settlement frequency	Annual, commencing at end of year 1
Fairfax pays to broker	Total return on Reston Industries stock
Broker pays to Fairfax	Total return on S&P 500 Stock Index

Table 2: Option Characteristics

	Reston	S&P 500
Stock price	$50.00	$1,400.00
Strike price	$50.00	$1,400.00
Interest rate	6.00%	6.00%
Dividend yield	0.00%	0.00%
Time to expiration (years)	0.5	0.5
Volatility	40.00%	17.00%
Beta Coefficient	1.23	1
Correlation	0.4	

Table 3: Regular and Exotic Options (Option Values)

	Reston	S&P 500
European call	$6.31	$6.31
European put	$4.83	$4.83
American call	$6.28	$6.28
American put	$4.96	$4.96

Table 4: Reston Stock Option Sensitivities

	Delta
European call	0.5977
European put	–0.4023
American call	0.5973
American put	–0.4258

Table 5: S&P 500 Option Sensitivities

	Delta
European call	0.622
European put	–0.378
American call	0.621
American put	–0.441

Potter has also been asked to evaluate the interest rate risk of an intermediate size bank. The bank has a large floating rate liability of $100,000,000 on which it pays the MRR on a quarterly basis. Potter is concerned about the significant interest rate risk the bank incurs because of this liability: since most of the bank's assets are invested in fixed rate instruments there is a considerable duration mismatch. Some of the bank's assets are floating rate notes tied to MRR, however, the total par value of these securities is significantly less than the liability position.

Potter considers both swaps and interest rate options. The interest rate options are 2-year caps and floors with quarterly exercise dates. Potter wishes to hedge the entire liability.

Potter has obtained the prices for an at-the-money 6 month cap and floor with quarterly exercise. These are shown in Table 6.

Table 6: At-the-Money 0.5 year Cap and Floor Values

Price of at-the-money Cap	$133,377
Price of at-the-money Floor	$258,510

Question #16 - 19 of 111

Question ID: 1473757

Williamson would like to consider neutralizing his Reston equity position from changes in Reston's stock price. Using the information in Tables 3 and 4 how many standard Reston European options would have to be bought/sold in order to create a delta neutral portfolio?

A) Sell 370,300 call options.

B) Sell 497,141 put options.

C) Buy 497,141 put options.

Question #17 - 19 of 111

Question ID: 1473758

Williamson is very interested in the total return swap. He asks Potter how much it would cost to enter into this transaction. Which of the following is the *most likely* cost of the swap at inception?

A) $45,007.

B) $0.

C) $340,885.

Question #18 - 19 of 111

Question ID: 1476157

Potter analyzes alternative hedging strategies to address the risk of the bank's large floating-rate liability. Which of the following is the *most appropriate* transaction to efficiently hedge the interest rate risk for the floating rate liability without sacrificing potential gains from interest rate decreases?

A) Sell an interest rate floor and buy an interest rate cap.

B) Sell an interest rate cap.

C) Buy an interest rate cap.

Question #19 - 19 of 111

Question ID: 1473760

Potter is now considering some of the bank's floating rate assets. Which of the following transactions is the *most appropriate* to minimize the interest rate risk of these assets without sacrificing upside gains?

A) Buy a floor.

B) Buy a cap.

C) Buy a collar.

Question #20 of 111

Question ID: 1473710

Regarding options on a stock without dividends, it is:

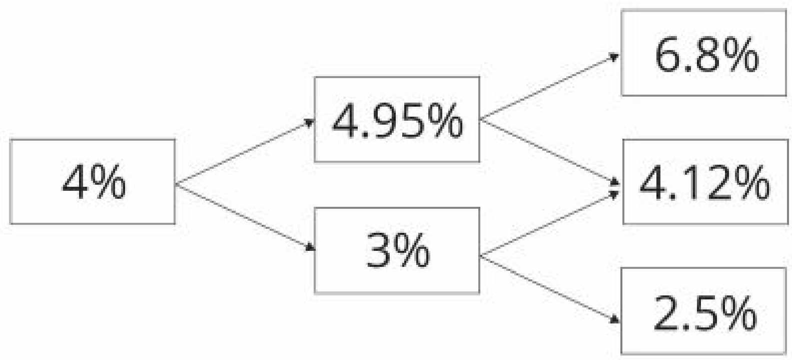
A) sometimes worthwhile to exercise calls early but not puts.

B) sometimes worthwhile to exercise puts early but not calls.

C) never worthwhile to exercise puts or calls early.

Lowell Wood is using the binomial option-pricing model to price interest rate options. She has obtained the following 2-year, annual rate tree (based on an assumed volatility of interest rates of 25%).

Exhibit 1



Wood has been asked help a colleague with the valuation of an interest rate put. The interest rate put option has 2 years to maturity and a strike price of 4.5% and is based on 360 day MRR. The option has a notional principal of $10m.

Wood has discovered that the Black model may be used to price options on interest rates by viewing the interest rate option as an option on a FRA. She is currently writing a research note for her team and makes the following three notes regarding the Black model:

Note 1:	"When using the Black model care needs to be taken to ensure that the payoff is discounted from the end of the borrowing and lending period (i.e., the maturity of the rate underlying the FRA), rather than the exercise date of the option."
Note 2:	"Given an interest rate option is an option on a FRA, call options will gain in value when interest rates rise and put options will fall in value."
Note 3:	"The accrual period needs to be factored in when valuing the option. This is because quoted rates are annual rates but in reality, the time between the FRA expiration and the maturity of borrowing and lending may not be one year. The accrual period can be viewed as a fraction of a year."

Wood asks for information about interest rate caps and floors. Newman makes the following comments:

Comment 1:	"A long FRA can be viewed as equivalent to a long interest rate call and a short interest rate put with the same strike and time to expiration."
Comment 2:	"Given a cap is a series of interest rate call options with identical strike prices and a floor is a series of interest rate put options also with identical strike prices, a short cap and long floor with identical strike prices would create a pay fixed receive floating interest rate swap."

Question #21 - 24 of 111

Question ID: 1478225

Using the information about the interest rate put and the spot and forward rates in Exhibit 1, which of the following is *closest* to the value of the put? Assume that the option cash settle at time 2.

A) $44,250.

B) $64,250.

C) $84,250.

Question #22 - 24 of 111

Question ID: 1473714

How many of Wood's notes regarding the Black model used to value interest options are correct?

A) All three notes are correct.

B) Only two of the notes are correct.

C) Only one of the notes is correct.

Question #23 - 24 of 111

Question ID: 1473715

Newman's Comment 1 is *best* described as:

A) correct.

B) incorrect as to the strike price of the options.

C) incorrect as to the equivalence to a long FRA.

Question #24 - 24 of 111

Question ID: 1473716

Newman's Comment 2 is *best* described as:

A) correct.

B) incorrect as buying a floor is not equivalent to buying interest rate put options.

C) incorrect as long floor, short cap would create a pay floating, receive fixed interest rate swap.

Question #25 of 111

Question ID: 1473735

Mark Roberts anticipates utilizing a floating rate line of credit in 90 days to purchase $10 million of raw materials. To get protection against any increase in the expected MRR yield curve, Roberts should:

A) write a receiver swaption.

B) buy a payer swaption.

C) buy a receiver swaption.

Question #26 of 111

Question ID: 1473734

A payer swaption gives its holder:

A) the right to enter a swap in the future as the floating-rate payer.

B) an obligation to enter a swap in the future as the fixed-rate payer.

C) the right to enter a swap in the future as the fixed-rate payer.

Question #27 of 111

Question ID: 1473703

Which of the following is NOT one of the assumptions of the Black-Scholes-Merton (BSM) option-pricing model?

A) There are no transaction costs, regulatory constraints, or taxes.

B) The options valued are European style (early exercise is not allowed).

C) The yield on the underlying has a known and constant volatility.

Question #28 of 111

Question ID: 1473687

Referring to put-call parity, which one of the following alternatives would allow you to create a synthetic European call option?

A) Sell the stock; buy a European put option on the same stock with the same exercise price and the same maturity; invest an amount equal to the present value of the exercise price in a pure-discount riskless bond.

B) Buy the stock; sell a European put option on the same stock with the same exercise price and the same maturity; short an amount equal to the present value of the exercise price worth of a pure-discount riskless bond.

C) Buy the stock; buy a European put option on the same stock with the same exercise price and the same maturity; short an amount equal to the present value of the exercise price worth of a pure-discount riskless bond.

Question #29 of 111

Question ID: 1473750

The value of a put option will be higher if, all else equal, the:

A) exercise price is lower.

B) underlying asset has positive cash flows.

C) underlying asset has less volatility.

Question #30 of 111

Question ID: 1473761

When an option's gamma is higher:

A) a delta hedge will be more effective.

B) delta will be higher.

C) a delta hedge will perform more poorly over time.

Question #31 of 111

Question ID: 1473749

Which of the following is the *best* approximation of the gamma of an option if its delta is equal to 0.6 when the price of the underlying security is 100 and 0.7 when the price of the underlying security is 110?

A) 1.00.

B) 0.01.

C) 0.10.

Question #32 of 111

Question ID: 1473763

Which of the following *best* explains the sensitivity of a call option's value to volatility? Call option values:

A) increase as the volatility of the underlying asset increases because call options have limited risk but unlimited upside potential.

B) are not affected by changes in the volatility of the underlying asset.

C) increase as the volatility of the underlying asset increases because investors are risk seekers.

Question #33 of 111

Question ID: 1473764

Which of the following *best* describes the implied volatility method for estimated volatility inputs for the Black-Scholes model? Implied volatility is found:

A) using the most current stock price data.

B) using historical stock price data.

C) by solving the Black-Scholes model for the volatility using market values for the stock price, exercise price, interest rate, time until expiration, and option price.

Question #34 of 111

Question ID: 1473708

Pete Jenkins makes the following statement about options:

"$N(d_2)$ is interpreted as the risk-neutral probability that a call option will expire in the money. Similarly, $N(-d_2)$ is the risk-neutral probability that a put option will expire in the money."

Jenkins is *most likely*:

A) incorrect about the risk-neutral probability of put option expiring in the money.

B) correct.

C) incorrect about the risk-neutral probability of call option expiring in the money.

Question #35 of 111

Question ID: 1473762

If we use four of the inputs into the Black-Scholes-Merton option-pricing model and solve for the asset price volatility that will make the model price equal to the market price of the option, we have found the:

A) historical volatility.

B) option volatility.

C) implied volatility.

Question #36 of 111

Question ID: 1473691

Combining a short position in a stock with a long position in a call option on the stock will produce a payoff pattern equivalent to a:

A) long position in a put option on the stock.

B) short position in a put option on the stock.

C) risk-free security.

Question #37 of 111

Question ID: 1473686

Referring to put-call parity, which one of the following alternatives would allow you to create a synthetic riskless pure-discount bond?

A) Sell a European put option; sell the same stock; buy a European call option.

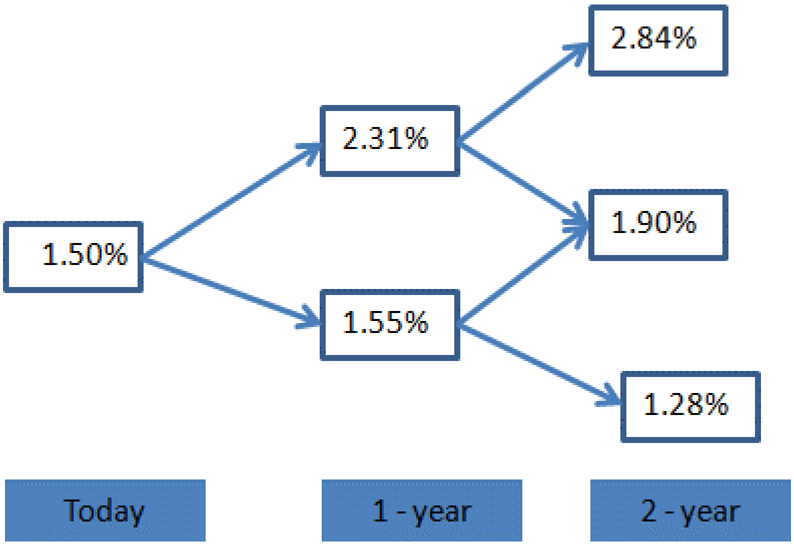
B) Buy a European put option; sell the same stock; sell a European call option.

C) Buy a European put option; buy the same stock; sell a European call option.

Question #38 of 111

Question ID: 1473697

Given the following interest rate tree:



The value of a 2-period European put option with strike rate of 2% and notional principal of $1 million is *closest* to:

A) $3,109

B) $2,230

C) $2,020

Question #39 of 111

Question ID: 1586298

Dividends on a stock can be incorporated into the valuation model of an option on the stock by:

A) subtracting the present value of the dividend from the current stock price.

B) adding the present value of the dividend to the current stock price.

C) subtracting the future value of the dividend from the current stock price.

Question #40 of 111

Question ID: 1473707

Bob Dilla, CFA makes the following statement about call options:

"Call options on stock can be thought of as leveraged stock investment where $N(d_1)$ units of stock is purchased using $e^{-rT}XN(-d_2)$ of borrowed funds."

Dilla is *most likely*:

A) correct.

B) incorrect about use of $e^{-rT}XN(-d_2)$ borrowed funds.

C) incorrect about $N(d_1)$ units of stock.

Question #41 of 111

Question ID: 1473751

Which of the following statements concerning vega is *most* accurate? Vega is greatest when an option is:

A) far out of the money.

B) far in the money.

C) at the money.

Question #42 of 111

Question ID: 1542002

A stock is priced at 40 and the periodic risk-free rate of interest is 8%. The value of a two-period European call option with a strike price of 37 on a share of stock using a binomial model with an up factor of 1.20 and down factor of 0.833 is *closest* to:

A) $9.13.

B) $9.25.

C) $3.57.

Question #43 of 111

Question ID: 1473693

Zetion Inc stock (current price $28) has 1-year call options with an exercise price of $30 trading at $2.07. The stock can increase by 15% or decrease by 13% over the next year and the risk-free rate is 3%. Arbitrage profits are *most likely*:

A) not possible.

B) possible by purchasing 28 shares and writing 100 calls.

C) possible by purchasing 100 calls and short selling 28 shares.

Ronald Franklin, CFA, has recently been promoted to junior portfolio manager for a large equity portfolio at Davidson-Sherman (DS), a large multinational investment-banking firm. He is specifically responsible for the development of a new investment strategy that DS wants all equity portfolio managers to implement. Upper management at DS has instructed its portfolio managers to begin overlaying option strategies on all equity portfolios. The relatively poor performance of many of their equity portfolios has been the main factor behind this decision. Prior to this new mandate, DS portfolio managers had been allowed to use options at their own discretion, and the results were somewhat inconsistent. Some portfolio managers were not comfortable with the most basic concepts of option valuation and their expected return profiles, and simply did not utilize options at all. Upper management of DS wants Franklin to develop an option strategy that would be applicable to all DS portfolios regardless of their underlying investment composition. Management views this new implementation of option strategies as an opportunity to either add value or reduce the risk of the portfolio.

Franklin gained experience with basic options strategies at his previous job. As an exercise, he decides to review the fundamentals of option valuation using a simple example. Franklin recognizes that the behavior of an option's value is dependent on many variables and decides to spend some time closely analyzing this behavior. His analysis has resulted in the information shown in Exhibit 1 and Exhibit 2 for European style options.

Exhibit 1: Input for European Options

Stock Price (S)	100
Strike Price (X)	100
Interest Rate (r)	0.07
Dividend Yield (q)	0.00
Time to Maturity (years) (t)	1.00
Volatility (Std. Dev.)(Sigma)	0.20

Black-Scholes Put Option Value	$4.7809

Exhibit 2: European Option Sensitivities

Sensitivity	Call	Put
Delta	0.6736	−0.3264
Gamma	0.0180	0.0180
Theta	−3.9797	2.5470
Vega	36.0527	36.0527
Rho	55.8230	−37.4164

Question #44 - 47 of 111

Question ID: 1586289

Using the information in Exhibit 1, Franklin wants to compute the value of the corresponding European call option. Which of the following is the *closest* to Franklin's answer?

A) $11.54.

B) $4.78.

C) $5.55.

Question #45 - 47 of 111

Question ID: 1586290

Franklin wants to know how the put option in Exhibit 1 behaves when all the parameters are held constant except the delta. Which of the following is the *best* estimate of the change in the put option's price when the underlying equity increases by $1?

A) −$0.37.

B) −$0.33.

C) −$3.61.

Question #46 - 47 of 111

Question ID: 1586291

Franklin computes the rate of change in the European put option delta value, given a $1 increase in the underlying equity. Using the information in Exhibit 1 and Exhibit 2, which of the following is the *closest* to Franklin's answer?

A) −0.3264.

B) 0.6736.

C) 0.0180.

Question #47 - 47 of 111

Question ID: 1586292

Franklin wants to know if the option sensitivities shown in Exhibit 2 have minimum or maximum bounds. Which of the following are the minimum and maximum bounds, respectively, for the put option delta?

A) −1 and 1.

B) There are no minimum or maximum bounds.

C) −1 and 0.

Question #48 of 111

Question ID: 1473732

Suppose a forward rate agreement (FRA) requires us to exchange six-month MRR one year from now for a fixed rate of interest of 8%. In other words, we will pay floating and receive fixed. Which of the following structures is *equivalent* to this FRA? A long:

A) call and a short put on MRR with a strike rate of 8% and six months to expiration.

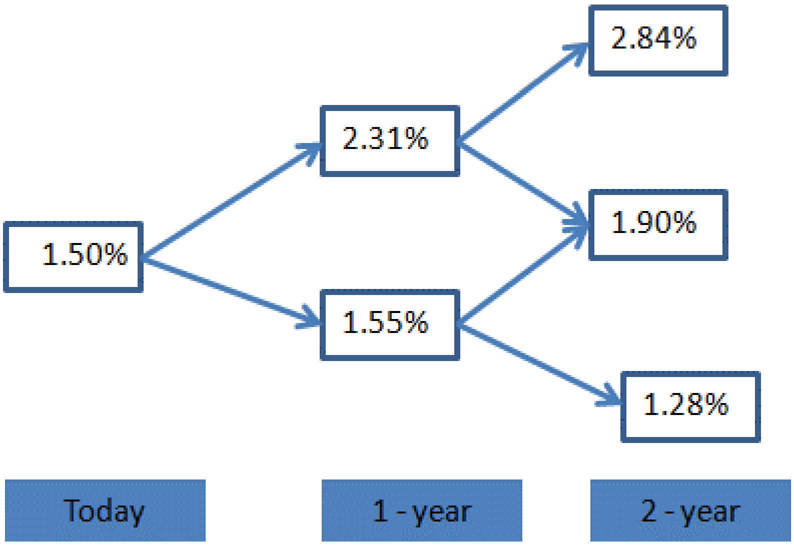
B) call and a short put on MRR with a strike rate of 8% and twelve months to expiration.

C) put and a short call on MRR with a strike rate of 8% and twelve months to expiration.

Question #49 of 111

Question ID: 1473696

Given the following interest rate tree:



The value of a 2-period European call option with strike rate of 2% and notional principal of $1 million is closest to:

A) $3,549

B) $2,022

C) $4,122

Question #50 of 111

Question ID: 1586299

Early exercise of in-the-money American options on:

A) futures is sometimes worthwhile but never is for options on forwards.

B) forwards is sometimes worthwhile but never is for options on futures.

C) both futures and forwards is sometimes worthwhile.

Question #51 of 111

Question ID: 1473739

A cap on a floating rate note, from the bondholder's perspective, is equivalent to:

A) owning a series of calls on fixed income securities.

B) writing a series of puts on fixed income securities.

C) writing a series of interest rate puts.

Jacob Bower is a bond strategist who would like to begin using fixed-income derivatives in his strategies. Bower has a firm understanding of the properties fixed-income securities. However, his understanding of interest rate derivatives is not nearly as strong. He decides to train himself on the valuation and sensitivity of interest rate derivatives using various interest rate scenarios. He considers the forward London Interbank Offered Rate (LIBOR) interest rate environment shown in Table 1. Using a rounded daycount (i.e., 0.25 years for each quarter) he has also computed the corresponding implied spot rates resulting from these LIBOR forward rates. These are included in Table 1.

Table 1: 90-Day LIBOR Forward Rates and Implied Spot Rates

Period (in months)	LIBOR Forward Rates	Implied Spot Rates
0 × 3	5.500%	5.5000%
3 × 6	5.750%	5.6250% •
6 × 9	6.000%	5.7499%
9 × 12	6.250%	5.8749%
12 × 15	7.000%	6.0997%
15 × 18	7.000%	6.2496%

Bower has also estimated the LIBOR forward rate volatilities to be 20%. The particular fixed instruments that Bower would like to examine are shown in Table 2. He also wants to analyze the strategy shown in Table 3.

Table 2: Interest Rate Instruments

Dollar Amount of Floating Rate Bond			$42,000,000
Floating Rate Bond paying LIBOR +			0.25%
Time to Maturity (years)			8
Cap Strike Rate		7.00%	
Floor Strike Rate		6.00%	

Interest Payments		quarterly	

Table 3: Initial Position in 90-day LIBOR Eurodollar Contracts

Contract Month (from now)	Strategy A (contracts)	Strategy B (contracts)
3 months	300	100
6 months	0	100
9 months	0	100

Question #52 - 55 of 111

Question ID: 1586306

Bower is a bit puzzled about how to use caps and floors. He wonders how he could benefit both from increasing and decreasing interest rates. Which of the following trades would *most likely* profit from this interest rate scenario?

A) Buy at the money cap and sell at the money floor.

B) Sell at the money cap and at the money floor.

C) Buy at the money cap and at the money floor.

Question #53 - 55 of 111

Question ID: 1586307

Bower has studied swaps extensively. However, he is not sure which of the following is the swap fixed rate for a one-year interest rate swap based on 90-day LIBOR with quarterly payments. Using the information in Table 1 and the formula below, what is the *most* appropriate swap fixed rate for this swap?

$$C = \frac{1 - Z_4}{Z_1 + Z_2 + Z_3 + Z_4}$$

where

$$Z_n = \frac{1}{1 + R_N} \text{price of n} - \text{zero} - \text{coupon bond per \$ of principal}$$

A) 5.75%.

B) 5.65%.

C) 6.01%.

Question #54 - 55 of 111

Question ID: 1586308

Bower computes the implied volatility of a one year caplet on the 90-day LIBOR forward rates to be 18.5%. Using the given information what does this mean for the caplet's market price relative to its theoretical price? The caplet's market price is:

A) overvalued.

B) undervalued.

C) undervalued or overvalued.

Question #55 - 55 of 111

Question ID: 1586309

For this question only, assume Bower expects the currently positively sloped LIBOR curve to shift upward in a parallel manner. Using a plain vanilla interest rate swap, which of the following will allow Bower to best take advantage of his expectations? Purchase a:

A) pay fixed interest rate swap.

B) floating rate bond and enter into a receive fixed swap.

C) receive fixed interest rate swap.

Question #56 of 111

Question ID: 1473733

The writer of a receiver swaption has:

A) an obligation to enter a swap in the future as the fixed-rate payer.

B) an obligation to enter a swap in the future as the floating-rate payer.

C) the right to enter a swap in the future as the floating-rate payer.

Nathan Detroit, a speculator, has come to you for technical advice regarding the pricing of swaps. He hopes to make big money in the swaps market from the exploitation of pricing discrepancies, but lacks an understanding of the principles underlying the pricing of swaps.

He asks you to consider a two-year, fixed-for-fixed, currency swap with semiannual payments. The domestic currency is the U.S. dollar and the foreign currency is the U.K. pound. The current exchange rate is $1.60 per pound. You forecast that the exchange rate would be $1.41 on the first settlement date. The notional principal of the swap is set at $10 million. The USD and £ term structure are shown in Exhibits 1 and 2 below.

Exhibit 1: Current USD Term Structure

Number of Days	MRR	Present Value Factor
180	0.0585	0.9716
360	0.0605	0.9430
540	0.0596	0.9179
720	0.0665	0.8826

Exhibit 2: Current £ Term Structure

Number of Days	MRR	Present Value Factor
180	0.0493	0.9759
360	0.0450	0.9569
540	0.0519	0.9278
720	0.0551	0.9007

Detroit has heard about the European put-call parity theorem and believes a synthetic call can be created through the use of a European put with the same strike as the call, a zero coupon bond with a face value equal to the strike price of the put and an underlying asset relating to the put and the call.

Detroit makes two comments regarding the BSM model and the Black model as follows:

Comment 1: $N(d_1)$ in the BSM is the probability that the option will expire in the money.

Comment 2: The probability that a receiver swaption will expire in the money is $N(-d_2)$.

Question #57 - 60 of 111

Question ID: 1473728

Calculate the USD swap fixed rate.

A) 3.16%.

B) 6.20%.

C) 6.32%.

Question #58 - 60 of 111

Question ID: 1473729

Using the information in Exhibits 1 and 2, which of the following is *closest* to the amount of the British pound paid on the first settlement date?

A) £165,000.

B) £187,234.

C) £264,000.

Question #59 - 60 of 111

Question ID: 1473730

Which of the following would create a synthetic call?

A) Buy the put and the underlying.

B) Sell the put, buy the underlying.

C) Sell the put and the underlying.

Question #60 - 60 of 111

Question ID: 1473731

How many of Nathan's comments are correct?

A) Neither comment is correct.

B) Only one statement is correct.

C) Both statements are correct.

Lucy Wang is the Chief Financial Officer of Sam Corporation. Sam Corporation has floating rate liabilities and wants to hedge against the possibility of rising interest rates. Wang is looking into using swaptions to hedge against interest rate risk.

The board of Sam Corporation are not familiar with both swaps and swaptions. To explain the characteristics of swaps, Wang explains to the board that swaps are similar to interest rate options.

Wang decides to use a swaption to hedge against interest rate risk. She knows that there are two types of swaptions just like there are both call and put options, and that the cash flows on swaps can be replicated using swaptions.

Lucy is very interested in the application of the Black model in pricing swaptions. After a quick search online she has found the following:

$$\text{pay} = (\text{AP})\text{PVA}\left[\left[\text{SFR} \times N(d_1) - X \times N(d_2)\right]\right] \times \text{NP}$$

where:

pay = value of the payer swaption

AP = 1 / number of settlement periods per year in the underlying swap

X = exercise rate specified in the swaption

NP = notional principal

Lucy is confused regarding what the notation PVA stands for and states:

"There are multiple payoffs on a swaption, each being the difference between the market swap rate at expiration and the exercise rate at each settlement date, over the swaps life. Given each payoff arises at a different point in time over the swaps life, each must be discounted back to the current period using discount rate specific to when it occurs. The PVA therefore must be an annuity factor summing all these specific discount rates."

Lucy also states:

"A payer swaption is right to enter a swap with a fixed rate equal to the strike price at the options maturity. The payoffs will therefore represent the difference in the swaptions strike price and the fixed market swap rate at the time of option expiry. Given we are comparing the payer swap underlying the swaption, versus the market rate of a payer swap at the option maturity, the floating payments can be ignored as they will offset. This is why the Black model only includes values for the current market swap fixed rate and fixed rate of the swap underlying the swaption (i.e., the strike)."

Question #61 - 64 of 111

Question ID: 1473718

Which of the following *best* describes how a payer swap could be replicated using a package of interest rate options?

A) The swap can be replicated by *buying* a package of interest rate *call* options *and* selling a package of interest rate put options at *different* strikes.

B) The swap can be replicated by *selling* a package of interest rate *call* options and *buying* a package of interest rate *put* options at the *same* strikes.

C) The swap can be replicated by *buying* a package of interest rate *call* options and *selling* a package of interest rate *put* options at the same strikes.

Question #62 - 64 of 111

Question ID: 1473719

Which of the following *best* describes how a payer swap could be replicated using interest rate swaptions?

A) The swap can be replicated by selling a payer swaption and buying a receiver swaption at the same strike.

B) The swap can be replicated by buying a payer swaption and selling a receiver swaption at the same strike.

C) The swap can be replicated by buying a payer swaption and selling a receiver swaption at different strikes.

Question #63 - 64 of 111

Question ID: 1473720

With regards to Lucy's comments regarding the PVA in the Black model and the offsetting of floating rates at expiry of the swaption, which of the following is *most* accurate?

A) Lucy is correct about the definition of the PVA and the offsetting of floating rates.

B) Lucy is correct about the offsetting of the floating rates but not the PVA.

C) Lucy is correct about the PVA but not the offsetting of floating rates.

Question #64 - 64 of 111

Question ID: 1473721

Which of the following comments relating to the Black model valuation of a swaption is the *most* accurate?

A) SFR in the formula identified by Lucy is the market swap fixed rate at the expiration of the swaption.

B) $N(d_2)$ is likely to be greater than $N(d_1)$.

C) A payer swaption will only be exercised in the market swap fixed rate at expiry is greater than the swaption's strike price.

Question #65 of 111

Question ID: 1473746

For a change in which of the following inputs into the Black-Scholes-Merton option pricing model will the direction of the change in a put's value and the direction of the change in a call's value be the same?

A) Volatility.

B) Exercise price.

C) Risk-free rate.

Question #66 of 111

Question ID: 1473709

Compared to the value of a call option on a stock with no dividends, a call option on an identical stock expected to pay a dividend during the term of the option will have a:

A) higher value only if it is an American style option.

B) lower value in all cases.

C) lower value only if it is an American style option.

Question #67 of 111

Question ID: 1473745

The value of a European call option on an asset with no cash flows is positively related to all of the following EXCEPT:

A) exercise price.

B) risk-free rate.

C) time to exercise.

Question #68 of 111

Question ID: 1473690

A non-dividend-paying option on a stock is *most likely* to be exercised early if the option is a(n):

A) European option.

B) call option.

C) put option.

Question #69 of 111

Question ID: 1473737

A floor on a floating rate note, from the bondholder's perspective, is equivalent to:

A) owning a series of puts on fixed income securities.

B) writing a series of interest rate puts.

C) owning a series of calls on fixed income securities.

Question #70 of 111

Question ID: 1473765

In order to compute the implied asset price volatility for a particular option, an investor:

A) must have the market price of the option.

B) must have a series of asset prices.

C) does not need to know the risk-free rate.

Susan Smith is analyzing the stock of FDL Inc. She has found the following quotations (all prices in dollars per share) on 91-day European options:

Exhibit 1

Option Strike Price	Call Premium	Put Premium
50	5.28	1.54
55	2.64	3.33
60	1.14	X
Other Information		
Risk-free interest rates	6%	
Annual volatility	30%	
FDL Inc share price	$53	

*FDL Inc currently does not pay dividends

Smith wants to value the equity call options of FDL Inc. using the Black-Scholes-Merton (BSM) model. She wants to understand the assumptions and the limitations of the model and asks David Wang for help. Wang provides the information shown in Exhibits 2 and 3.

Exhibit 2

Assumptions of BSM Model

Assumption 1

The underlying asset returns follow a normal distribution

Assumption 2

Options are European style

Exhibit 3

Assumptions of BSM Model	Implications
The risk-free rate is known and constant over the options life	**Implication 1** Useful for pricing options on bond prices and interest rates
The continuously compounded yield on the asset is constant	**Implication 2** BSM model can be modified to account for cash flows on the underlying

Question #71 - 74 of 111

Question ID: 1473699

Using information in Exhibit 1, the value of $60 strike put option is *closest* to:

A) $4.99.

B) $5.86.

C) $7.27.

Question #72 - 74 of 111

Question ID: 1473700

Susan discovers that the fair value for the $55 strike put is in fact $3.85. Which of the following is the *most* appropriate set of transactions to exploit the mispricing (ignore transaction costs)?

A) Write a call option, buy a put option, buy one share, borrow the PV of strike.

B) Buy a call option, write a put option, buy one share, borrow the PV of strike.

C) Buy a call option, write a put option, sell short one share, put on deposit the PV of strike.

Question #73 - 74 of 111

Question ID: 1477072

Using information in Exhibit 2, which of the following statements about assumptions 1 and 2 is *most* accurate?

A) Both assumptions are correct.

B) Only one of the two assumptions is correct.

C) Both assumptions are incorrect.

Question #74 - 74 of 111

Question ID: 1542004

Using Exhibit 3, which of the following statements about implications 1 and 2 is *most* accurate?

A) Both implications are correct.

B) Both implications are incorrect.

C) Only one of the two implications is correct.

Rachel Barlow is a recent graduate of Columbia University with a Bachelor's degree in finance. She has accepted a position at a large investment bank, but first must complete an intensive training program to gain experience in several of the investment bank's areas of operations. Currently, she is spending three months at her firm's Derivatives Trading desk. One of the traders, Jason Coleman, CFA, is acting as her mentor, and will be giving her various assignments over the three month period.

One of the first projects Coleman asks Barlow to do is to compare different option trading strategies. Coleman would like Barlow to pay particular attention to strategy costs and their potential payoffs. Barlow is not very comfortable with option models, and knows she needs to be able to fully understand the most basic concepts in order to move on. She decides that she must first investigate how to properly price European and American style equity options.

Coleman has given Barlow software that provides a variety of analytical information using three valuation approaches: the Black-Scholes model, the Binomial model, and Monte Carlo simulation. Barlow has decided to begin her analysis using a variety of different scenarios to evaluate option behavior. The data she will be using in her scenarios is provided in Exhibits 1 and 2. Note that all of the rates and yields are on a continuous compounding basis.

Exhibit 1

Stock Price (S)	\$100.00
Strike Price (X)	\$100.00
Interest Rate (r)	7.0%
Dividend Yield (q)	0.0%
Time to Maturity (years)	0.5
Volatility (Std. Dev.)	20.0%
Value of Put	\$3.9890

Exhibit 2

Stock Price (S)	\$110.00
Strike Price (X)	\$100.00
Interest Rate (r)	7.0%
Dividend Yield (q)	0.0%
Time to Maturity (years)	0.5
Volatility (Std. Dev.)	20.0%
Value of Call	\$14.8445
$N(d_1)$	0.8394
$N(d_2)$	0.8025

Exhibit 3

Stock Price (S)	\$115.00
Strike Price (X)	\$100.00
Interest Rate (r)	7.0%
Dividend Yield (q)	0.0%

Time to Maturity (years)	0.5
Volatility (Std. Dev.)	20.0%
Value of Call	$19.2147
Value of Put	$0.7753

Question #75 - 78 of 111

Question ID: 1586294

Barlow notices that the stock in Exhibit 1 does not pay dividends. If the stock begins to pay a dividend, how will the price of a call option on that stock be affected? The price of the call option:

A) may either increase or decrease.

B) will decrease.

C) will increase.

Question #76 - 78 of 111

Question ID: 1586295

Barlow calculated the value of an American call option on the stock shown in Exhibit 2. Which of the following is *closest* to the value of this call option?

A) $15.41.

B) $14.84.

C) $15.12.

Question #77 - 78 of 111

Question ID: 1586296

Using the information in Exhibit 2, Barlow computes the value of a European put option. Which of the following is *closest* to the value of this option?

A) $1.97.

B) $4.84.

C) $1.41.

Question #78 - 78 of 111

Question ID: 1586297

Barlow notices that the stock in Exhibit 2 does not pay dividends. If the stock starts to pay a dividend, how will the price of a put option on that stock be affected?

A) Increase.

B) Decrease.

C) Increase or decrease.

Question #79 of 111

Question ID: 1473692

To create a synthetic short position in a stock, an investor can buy:

A) a call option on the stock and sell a put option on the stock.

B) a put option on the stock and sell a call option on the stock.

C) both a call option on the stock and a put option on the stock.

Question #80 of 111

Question ID: 1473704

Which of the following is *least likely* one of the assumptions of the Black-Scholes-Merton option pricing model?

A) Changes in volatility are known and predictable.

B) The risk-free rate of interest is known and does not change over the term of the option.

C) The options are European.

Question #81 of 111

Question ID: 1473694

DTK Inc stock (current price \$55) has 1-year call options with an exercise price of \$55 trading at \$4.92. The stock can increase by 20% or decrease by 15% over the next year and the risk-free rate is 5%. Arbitrage profits are *most likely*:

A) not possible.

B) possible by purchasing 100 calls and short selling 57 shares.

C) possible by purchasing 57 shares and writing 100 calls.

Question #82 of 111

Question ID: 1473741

Which of the following *best* represents an interest floor?

A) A portfolio of put options on an interest rate.

B) A put option on an interest rate.

C) A portfolio of call options on an interest rate.

Joel Franklin, CFA, has recently been promoted to junior portfolio manager for a large equity portfolio at Davidson Sherman (DS), a large multinational investment banking firm. The portfolio is subdivided into several smaller portfolios. In general, the portfolios are composed of U.S. based equities, ranging from medium to large-cap stocks. Currently, DS is not involved in any foreign markets. In his new position, he will now be responsible for the development of a new investment strategy that DS wants all of its equity portfolios to implement. The strategy involves overlaying option strategies on its equity portfolios. Recent performance of many of their equity portfolios has been poor relative to their peer group. The upper management at DS views the new option strategies as an opportunity to either add value or reduce risk.

Franklin recognizes that the behavior of an option's value is dependent upon many variables and decides to spend some time closely analyzing this behavior. He took an options strategies class in graduate school a few years ago, and feels that he is fairly knowledgeable about the valuation of options using the Black-Scholes model. Franklin understands that the volatility of the underlying asset returns is one of the most important contributors to option value. Therefore, he would like to know when the volatility has the largest effect on option value. Upper management at DS has also requested that he further explore the concept of a delta neutral portfolio. He must determine how to create a delta neutral portfolio, and how it would be expected to perform under a variety of scenarios. Franklin is also examining the change in the call option's delta as the underlying equity value changes. He also wants to determine the minimum and maximum bounds on the call option delta. Franklin has been authorized to purchase calls or puts on the equities in the portfolio. He may not, however, establish any uncovered or "naked" option positions. His analysis has resulted in the information shown in Exhibit 1 and Exhibit 2 for European style options.

Exhibit 1: Input for European Options

Stock Price (S)	100
Strike Price (X)	100
Interest Rate (r)	0.07
Dividend Yield (q)	0
Time to Maturity (years) (t)	1
Volatility (Std. Dev.) (sigma)	0.2
Black-Scholes Put Option Value	$4.7809

Exhibit 2: European Option Sensitivities

Sensitivity	Call	Put
Delta	0.6736	−0.3264
Gamma	0.0180	0.0180
Theta	−3.9797	2.5470
Vega	36.0527	36.0527
Rho	55.8230	−37.4164

Question #83 - 86 of 111

Question ID: 1586321

Which of the following *most* accurately describes when the call option delta reaches its minimum bound? The call option reaches its minimum bound when call option is:

A) at the money.

B) the option's delta has no minimum bound.

C) far out of the money.

Question #84 - 86 of 111

Question ID: 1586322

If the portfolio has 10,000 shares of the underlying stock and he wants to completely hedge the price risk using options, what kind of options should Franklin buy?

A) Call and put options.

B) Call options.

C) Put options.

Question #85 - 86 of 111

Question ID: 1586323

Compute the number of shares of stock necessary to create a delta neutral portfolio consisting of 100 long put options in Exhibit 2 and the stock.

A) 67.36.

B) 32.64.

C) −32.64.

Question #86 - 86 of 111

Question ID: 1586324

Compute the number of shares of stock necessary to create a delta neutral portfolio consisting of 100 long call options in Exhibit 2 and the stock.

A) 67.36.

B) −67.36.

C) −32.64.

Question #87 of 111

Question ID: 1473742

Suppose a forward rate agreement (FRA) calls for us to receive the six-month MRR two years from now for a payment of a fixed rate of interest of 6%. Which of the following structures is equivalent to this long FRA? A long:

A) put and a short call on MRR with a strike rate of 6% and two years to expiration.

B) call and a short put on MRR with a strike rate of 6% and two years to expiration.

C) call on MRR with a strike rate of 6% and eighteen months to expiration.

Al Bingly, CFA, is a derivatives specialist who attempts to identify and make short-term gains from trading mispriced options. One of the strategies that Bingly uses is to look for arbitrage opportunities in the market for European options. This strategy involves creating a synthetic call from other instruments at a cost less than the market value of the call itself, and then selling the call. During the course of his research, he observes that Hilland Corporation's stock is currently priced at $56, while a European-style put option with a strike price of $55 is trading at $0.40 and a European-style call option with the same strike price is trading at $2.50. Both options have 6 months remaining until expiration. The risk-free rate is currently 4 percent.

Bingly often uses the binomial model to estimate the fair price of an option. He then compares his estimated price to the market price. He observes that Dale Corporation's stock has a current market price of $200, and he predicts that its price will either be $166.67 or $240 in one year. The risk-free rate is currently 4 percent. He also observes that the price of a one-year call with a $220 strike price is $11.11.

Bingly also uses the Black-Scholes-Merton model to price options. His stated rationale for using this model is that he believes the prices of the stocks he analyzes follow a lognormal distribution, and because the model allows for a varying risk-free rate over the life of the option. His plan is to use a statistical technique to estimate the volatility of a stock, enter it into the Black-Scholes-Merton model, and see if the associated price is higher or lower than the observed market price of the options on the stock.

Bingly wishes to apply the Black-Scholes-Merton model to both non-dividend paying and dividend paying stocks. He investigates how the presence of dividends will affect the estimated call and put price.

Question #88 - 91 of 111

Question ID: 1586301

The one-year call option on Dale Corporation:

A) is underpriced.

B) is overpriced.

C) may be over or underpriced. The given information is not sufficient to give an answer.

Question #89 - 91 of 111

Question ID: 1586302

Bingly's sentiments towards the Black-Scholes-Merton (BSM) model regarding a lognormal distribution of prices and a variable risk-free rate are:

A) correct for both reasons.

B) correct concerning the distribution of stocks but incorrect concerning the risk-free rate.

C) incorrect for both reasons.

Question #90 - 91 of 111

Question ID: 1586303

If Bingly forecasts the volatility for a stock and find that it is significantly greater than that implied by the prices of the puts and calls of the stock, he would conclude that:

A) the puts are overpriced and the calls are underpriced.

B) puts and calls are underpriced.

C) puts and calls are overpriced.

Question #91 - 91 of 111

Question ID: 1586304

All else being equal, the greater the dividend paid by a stock the:

A) lower the call price and the higher the put price.

B) lower the call price and the lower the put price.

C) higher the call price and the lower the put price.

John Fairfax is a recently retired executive from Reston Industries. Over the years he has accumulated $10 million worth of Reston stock and another $2 million in a cash savings account. He hires Richard Potter, CFA, a financial adviser from Stan Morgan, LLC, to help him develop investment strategies. Potter suggests a number of interesting investment strategies for Fairfax's portfolio. Many of the strategies include the use of various equity derivatives. Potter explains to Fairfax that there are numerous options available for him to obtain almost any risk return profile he might need. Potter suggests that Fairfax consider options on both Reston stock and the S&P 500. Potter collects the information needed to evaluate options for each security. These results are presented in Table 1.

Table 1: Option Characteristics

	Reston	S&P 500
Stock price	$50.00	$1,400.00
Strike price	$50.00	$1,400.00
Interest rate	6.00%	6.00%
Dividend yield	0.00%	0.00%
Time to expiration (years)	0.5	0.5
Volatility	40.00%	17.00%
Beta Coefficient	1.23	1
Correlation	0.4	

Potter presents Fairfax with the prices of various options as shown in Table 2. Table 2 details standard European calls and put options. Potter presents the option sensitivities in Potter presents Fairfax with the prices of various options as shown in Table 3and Potter presents Fairfax with the prices of various options as shown in Table 4.

Table 2: Regular and Options (Option Values)

	Reston	S&P 500
European call	$6.31	$6.31
European put	$4.83	$4.83
American call	$6.28	$6.28
American put	$4.96	$4.96

Table 3: Reston Stock Option Sensitivities

	Delta
European call	0.5977
European put	−0.4023
American call	0.5973
American put	−0.4258

Table 4: S&P 500 Option Sensitivities

	Delta
European call	0.622
European put	−0.378
American call	0.621
American put	−0.441

Question #92 - 95 of 111

Question ID: 1586331

Given the information regarding the various Reston stock options, which option will increase the *most* relative to an increase in the underlying Reston stock price?

A) European call.

B) American put.

C) American call.

Question #93 - 95 of 111

Question ID: 1586332

Fairfax would like to consider neutralizing his Reston equity position from changes in the stock price of Reston. Using the information in Table 3 how many standard Reston European options would have to be either bought or sold in order to create a delta neutral portfolio?

A) Sell 334,616 put options.

B) Buy 300,703 put options.

C) Sell 334,616 call options.

Question #94 - 95 of 111

Question ID: 1586333

Fairfax remembers Potter explaining something about how options are not like futures and swaps because their risk-return profiles are non-linear. Which of the following option sensitivity measures does Fairfax need to consider to completely hedge his equity position in Reston from changes in the price of Reston stock?

A) Gamma and Theta.

B) Delta and Gamma.

C) Delta and Vega.

Question #95 - 95 of 111

Question ID: 1586334

Fairfax has heard people talking about "making a portfolio delta neutral." What does it mean to make a portfolio delta neutral? The portfolio:

A) is insensitive to stock price changes.

B) is insensitive to interest rate changes.

C) is insensitive to volatility changes in the returns on the underlying equity.

Gina Davalos, CFA is a portfolio manager for the Herron Investments. She is interested in hedging the equity risk of one of her clients, Lou Gier. Gier has 200,000 shares of a stock with the symbol QJX that he believes could take a dive in the next 9 months. Davalos gathers the following information to suggest potential strategies to offset the potential loss. Each option contract is for 100 options.

General Information:

QJX Current Stock Price	\$100.00
Risk-free rate	5.0%
QJX Dividend Yield	0.0%
Time to Maturity (years)	0.75

Option Information:

Strike Price	\$100.00
Value of Call	\$12.09
Delta on Call Option	0.6081
Value of Put (years)	\$8.41

Equity Swap Information:

Terms	9 months
Settlement frequency	Quarterly
Fixed rate	6.0%
Return on QJX	Variable

Futures Information:

Terms	9 months
Current Futures Price	$105.50

Question #96 - 99 of 111

Question ID: 1586326

In order to create a delta-neutral hedge using put option contracts, Davalos would *most accurately* need to:

A) buy 2,000 contracts.

B) buy 5,103 contracts.

C) sell 510,271 contracts.

Question #97 - 99 of 111

Question ID: 1586327

An equity swap to hedge the equity risk for Gier would result in a net return on the portfolio of a:

A) fixed rate of 4.5% for the year.

B) variable rate based on the total return of QJX stock.

C) fixed rate of 1.5% per quarter.

Question #98 - 99 of 111

Question ID: 1586328

If the equity swap is implemented and after 3 months the stock price has increased to $106.00, the net cash flow for the swap is:

A) a loss of $900,000.

B) a gain of $900,000.

C) zero.

Question #99 - 99 of 111

Question ID: 1586329

Based on the futures information, an arbitrage opportunity can be exploited by:

A) buying the stock QJX, and selling the futures.

B) buying the futures and buying the stock QJX.

C) selling the stock QJX and buying the futures.

Question #100 of 111

Question ID: 1534580

A stock is priced at 38 and the periodic risk-free rate of interest is 6%. What is the value of a two-period European put option with a strike price of 35 on a share of stock using a binomial model with an up factor of 1.15, a down factor of 0.87 and a risk-neutral up probability of 68%?

A) $0.57.

B) $0.64.

C) $2.58.

Mark Washington, CFA, is an analyst with BIC, a Bermuda-based investment company that does business primarily in the U.S. and Canada. BIC has approximately $200 million of assets under management, the bulk of which is invested in U.S. equities. BIC has outperformed its target benchmark for eight of the past ten years, and has consistently been in the top quartile of performance when compared with its peer investment companies. Washington is a part of the Liability Management group that is responsible for hedging the equity portfolios under management. The Liability Management group has been authorized to use calls or puts on the underlying equities in the portfolio when appropriate, in order to minimize their exposure to market volatility. They also may utilize an options strategy in order to generate additional returns.

One year ago, BIC analysts predicted that the U.S. equity market would most likely experience a slight downturn due to inflationary pressures. The analysts forecast a decrease in equity values of between 3 to 5% over the upcoming year and one-half. Based upon that prediction, the Liability Management group was instructed to utilize calls and puts to construct a delta-neutral portfolio. Washington immediately established option positions that he believed would hedge the underlying portfolio against the impending market decline.

As predicted, the U.S. equity markets did indeed experience a downturn of approximately 4% over a twelve-month period. However, portfolio performance for BIC during those twelve months was disappointing. The performance of the BIC portfolio lagged that of its peer group by nearly 10%. Upper management believes that a major factor in the portfolio's underperformance was the option strategy utilized by Washington and the Liability Management group. Management has decided that the Liability Management group did not properly execute a delta-neutral strategy. Washington and his group have been told to review their options strategy to determine why the hedged portfolio did not perform as expected. Washington has decided to undertake a review of the most basic option concepts, and explore such elementary topics as option valuation, an option's delta, and the expected performance of options under varying scenarios. He is going to examine all facets of a delta-neutral portfolio: how to construct one, how to determine the expected results, and when to use one. Management has given Washington and his group one week to immerse themselves in options theory, review the basic concepts, and then to present their findings as to why the portfolio did not perform as expected.

Question #101 - 104 of 111

Question ID: 1586316

Which of the following *best* explains a delta-neutral portfolio? A delta-neutral portfolio is perfectly hedged against:

A) small price decreases in the underlying asset.

B) all price changes in the underlying asset.

C) small price changes in the underlying asset.

Question #102 - 104 of 111

Question ID: 1586317

After discussing the concept of a delta-neutral portfolio, Washington determines that he needs to further explain the concept of delta. Washington draws the payoff diagram for an option as a function of the underlying stock price. Using this diagram, how is delta interpreted? Delta is the:

A) slope in the option price diagram.

B) level in the option price diagram.

C) curvature of the option price graph.

Question #103 - 104 of 111

Question ID: 1586318

Washington is trying to determine the value of a call option. When the slope of the at expiration curve is close to zero, the call option is:

A) in-the-money.

B) at-the-money.

C) out-of-the-money.

Question #104 - 104 of 111

Question ID: 1586319

BIC owns 51,750 shares of Smith & Oates. The shares are currently priced at $69. A call option on Smith & Oates with a strike price of $70 is selling at $3.50, and has a delta of 0.69 What is the number of call options necessary to create a delta-neutral hedge?

A) 0.

B) 75,000.

C) 14,785.

Question #105 of 111

Question ID: 1473740

To the issuer of a floating rate note, a cap is equivalent to:

A) writing a series of interest rate calls.

B) owning a series of interest rate calls.

C) owning a series of calls on a fixed income security.

Question #106 of 111

Question ID: 1473706

A bond analyst decides to use the BSM model to price options on bond prices. This model will *most likely* be inadequate because:

A) BSM cannot be modified to deal with cash flows like coupon payments.

B) the price of the underlying asset follows a lognormal distribution.

C) the risk free rate must be constant and known.

Question #107 of 111

Question ID: 1473736

Cal Smart wrote a 90-day receiver swaption on a 1-year MRR-based semiannual-pay $10 million swap with an exercise rate of 3.8%. At expiration, the market rate and MRR yield curve are:

Fixed rate 3.763%

180-days 3.6%

360-days 3.8%

The payoff to the writer of the receiver swaption at expiration is *closest* to:

A) $0.

B) -$3,600.

C) $3,600.

Question #108 of 111

Question ID: 1473744

Which of the following option sensitivities measures the change in the price of the option with respect to a decrease in the time to expiration?

A) Gamma.

B) Delta.

C) Theta.

Question #109 of 111

Question ID: 1473738

Which of the following *best* describes an interest rate cap? An interest rate cap is a package or portfolio of interest rate options that provide a positive payoff to the buyer if the:

A) T-Bond futures exceeds the strike price.

B) reference rate exceeds the strike rate.

C) reference rate is below the strike rate.

Question #110 of 111

Question ID: 1473754

The price of a June call option with an exercise price of $50 falls by $0.50 when the underlying non-dividend paying stock price falls by $2.00. The delta of a June put option with an exercise price of $50 *closest* to:

A) –0.25.

B) 0.25.

C) –0.75.

Question #111 of 111

Question ID: 1473747

The delta of an option is equal to the:

A) dollar change in the stock price divided by the dollar change in the option price.

B) dollar change in the option price divided by the dollar change in the stock price.

C) percentage change in option price divided by the percentage change in the asset price.